



TEKATHON-5.0-2026



ChainTrace

- **PROBLEM STATEMENT ID** - 26183
- **PROBLEM STATEMENT TITLE** - Real-Time Identification of Fraud-Linked Cryptocurrency Exchanges from Victim-Reported Suspect Wallet Addresses through Automated Blockchain Analysis.
- **THEME** - Cybersecurity × Blockchain — Dark Tech
- **PS CATEGORY** - SOFTWARE
- **TEAM NAME** - THE ELEMENTS



Chain Trace



A PLATFORM TO TRACK ILLICIT CRYPTO FLOW, FLAG FRAUDULENT ACCOUNTS & ANALYZE SUSPICIOUS FINANCIAL TRANSACTIONS

Investigators can see transactions — but not the complete financial story.

Crypto activity is pseudonymous, borderless and fast-moving; the evidence needed to connect a wallet to a real-world entity is often distributed across different data owners and systems.

\$154B

Illicit addresses received at least this value in 2025

162% YoY increase • lower-bound estimate

\$17B

Estimated stolen in crypto scams & fraud in 2025

AI-enabled scams were 4.5× more profitable

84%

Of illicit transaction volume involved stablecoins in 2025

Cross-border transferability is a major risk factor

Source: Chainalysis, 2026 Crypto Crime Report.

WHY THIS IS DIFFICULT TO TRACE

- **PSEUDONYMITY** Addresses and transaction history are visible, but real-world attribution requires additional lawful evidence.
- **OBFUSCATION** Layering, rapid multi-hop movement and cross-chain activity can fragment the observable fund trail.
- **SPEED** Funds can move through several counterparties while conventional investigation workflows are still gathering context.
- **JURISDICTION** A single flow can cross platforms, service providers and jurisdictions with different data-access pathways.

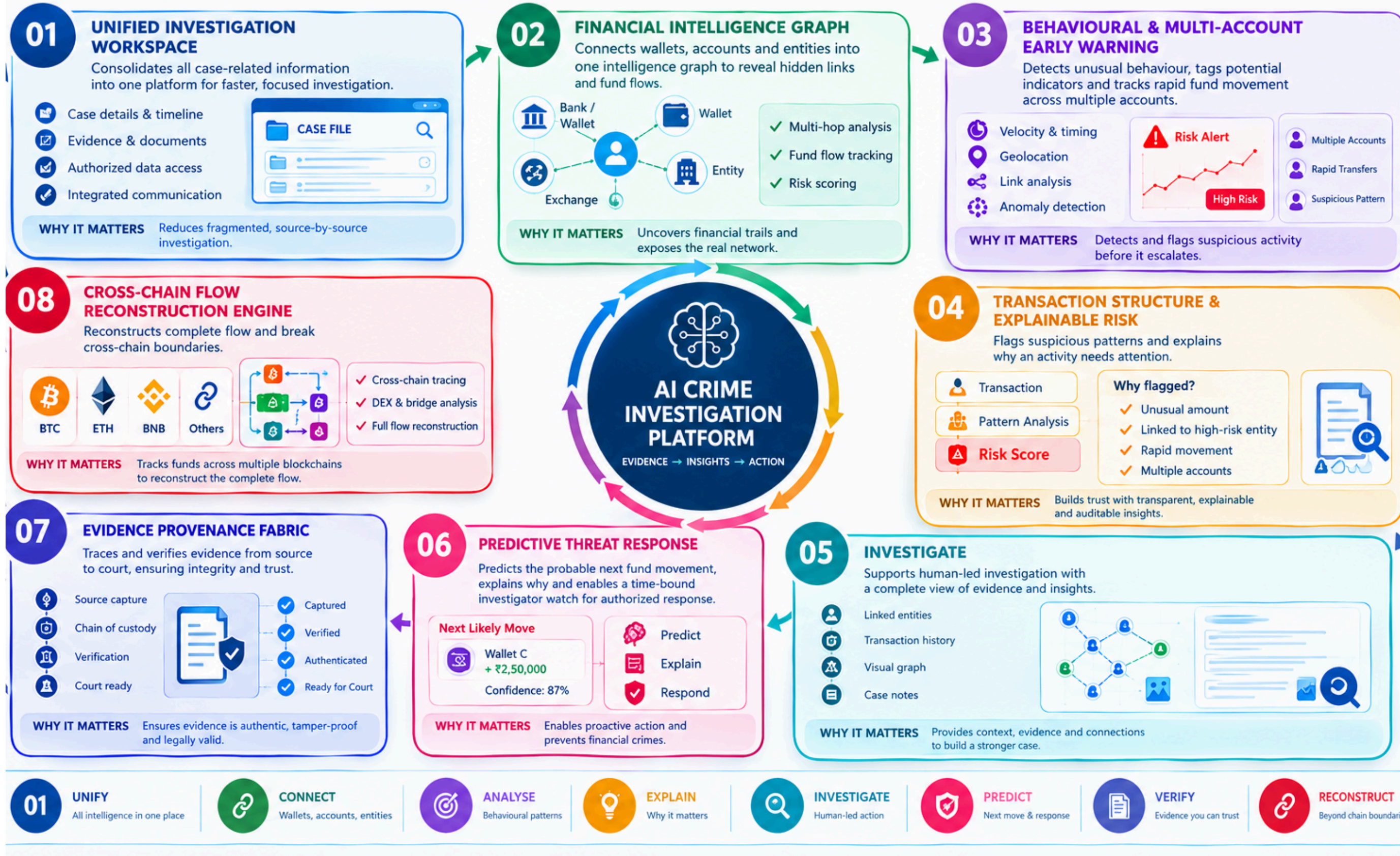
WHY CURRENT WORKFLOWS FALL SHORT

- **FRAGMENTED DATA** Blockchain, exchanges, financial records, STRs, sanctions data and OSINT sit in separate workflows.
- **MANUAL CORRELATION** Investigators must pivot between wallets, transaction hashes, accounts, cases and external sources.
- **LOW-CONTEXT ALERTS** A risk score without observable contributors does not explain what should be investigated next.
- **REACTIVE REVIEW** The system needs earlier prioritization while retaining source provenance, human validation and auditability.

THE INVESTIGATION GAP → FRAGMENTED TRANSACTION SIGNALS → NO SINGLE, EXPLAINABLE VIEW OF THE COMPLETE FINANCIAL NETWORK



TECHNICAL APPROACH





FREASIBILITY & VIABILITY

01 Technical Feasibility

Built using mature, production ready technologies

AI/ML: Graph ML, anomaly detection, explainable AI

Blockchain: Ethereum, Bitcoin, BNBchain, transaction indexing.

Backend: Python, FastAPI, Neo4j, Graph database.

Integration: Secure REST APIs with authorized institutions

WHY IT MATTERS No new blockchain infrastructure required.
Works with publicly available blockchain data plus authorized institutional intelligence

02 Operational Viability

Designed for investigators with explainability, auditability, and legal compliance.

Evidence provenance: every insight linked to its source, timestamps and integrity hash.

Human-in-the-loop: AI assists investigators; final decisions remain authorized by officials

WHY IT MATTERS Maintain chain of custody for digital evidence.
Investigator ready reports for review and court process

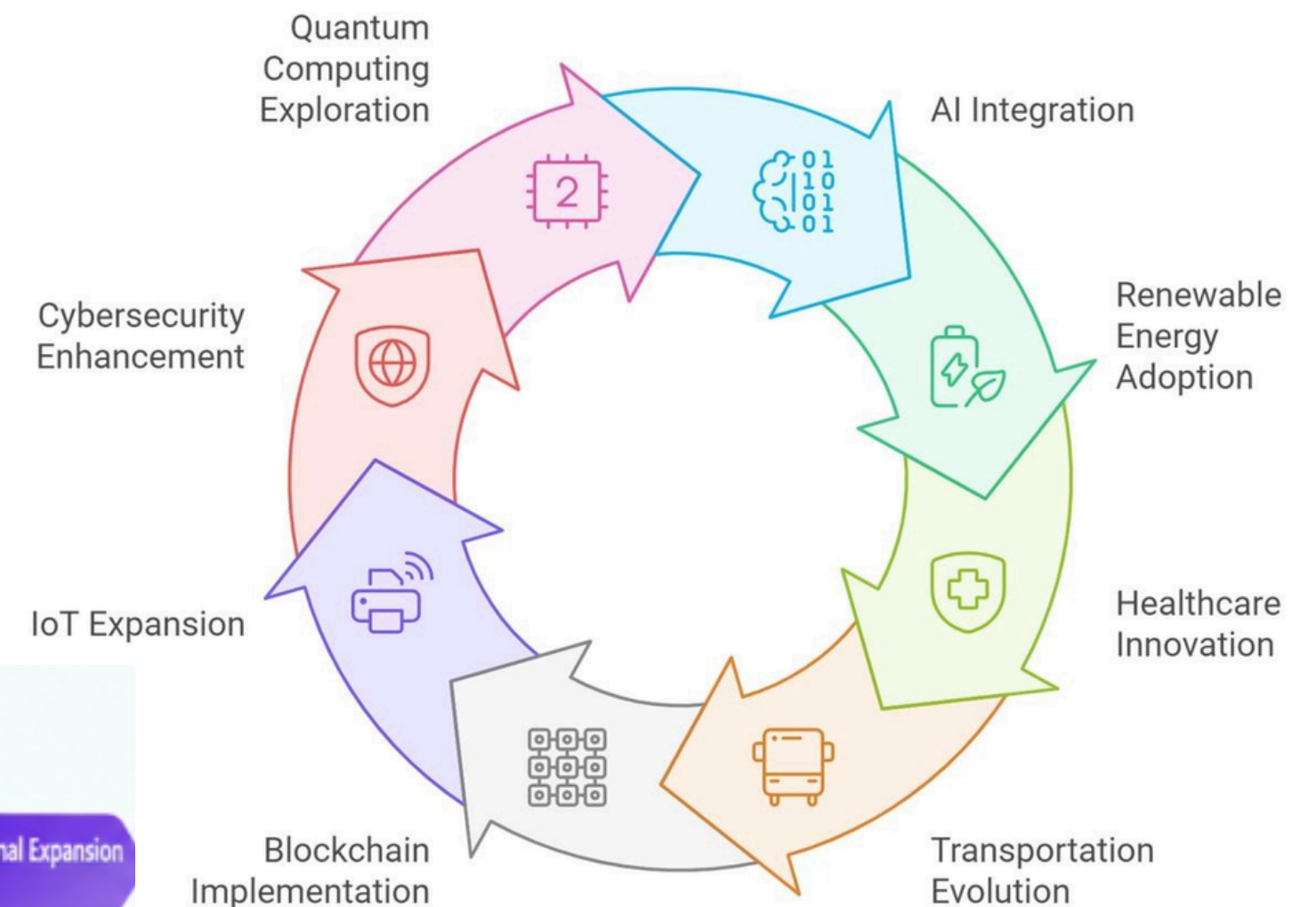
✓ FEASIBILITY & VIABILITY

Technically Feasible

Operationally Viable

Scalable for Law Enforcement

CHAINTRACE is designed to work with existing investigation infrastructure rather than replacing it. It integrates authorized intelligence sources, AI/ML analytics, and blockchain transaction intelligence into one explainable investigation platform.



03



DEPLOYMENT VIABILITY

Four-phase implementation roadmap



Phase 1 – Pilot
0 – 3 Months

- Chandigarh Police Cyber Cell
- Blockchain transaction monitoring



Phase 2 – Integration
3 – 9 Months

- Connect banks, exchanges,
- FIU/STR APIs & case management



Phase 3 – State Scale
9 – 18 Months

- Multi-department deployment
- Across Punjab cyber units

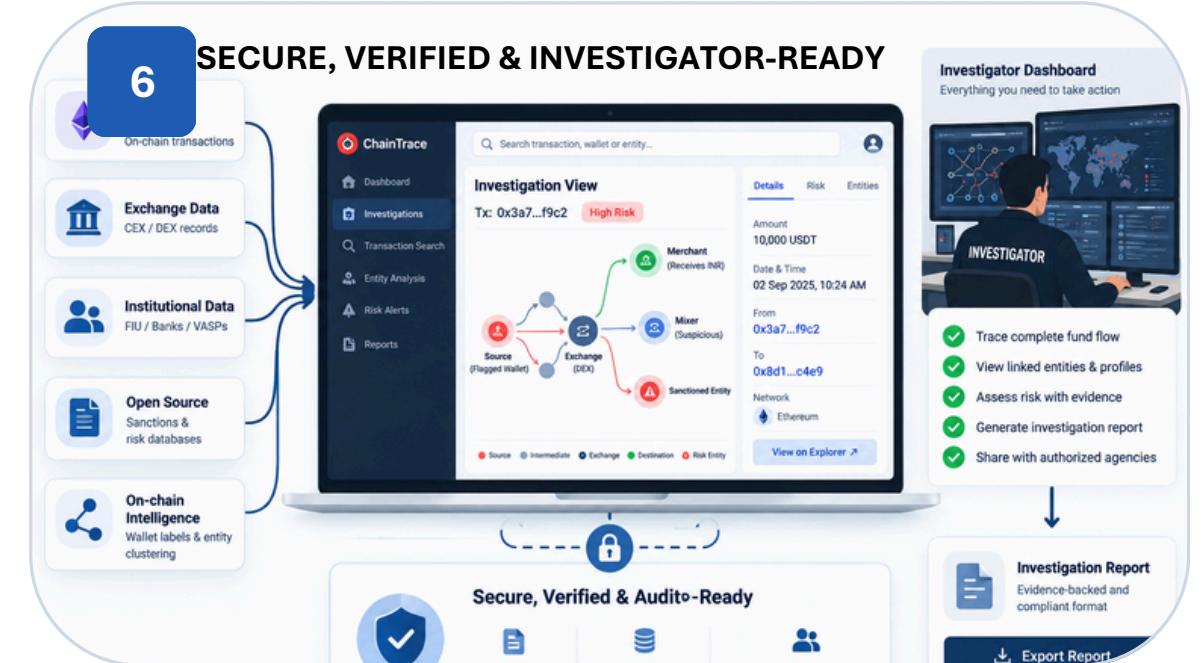
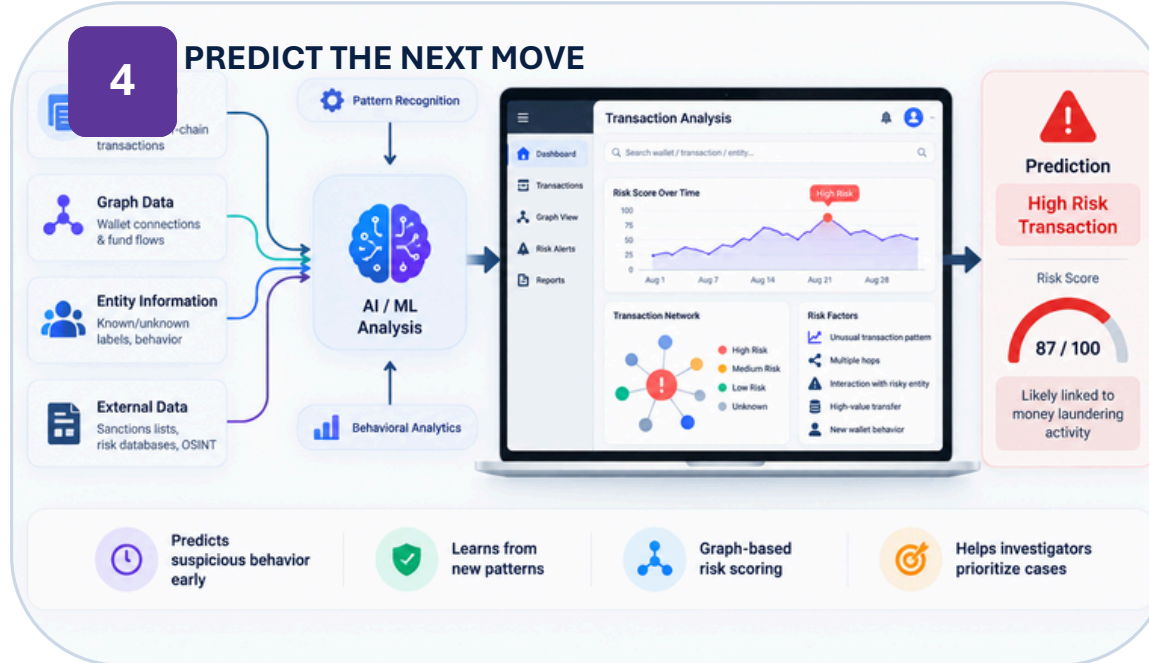
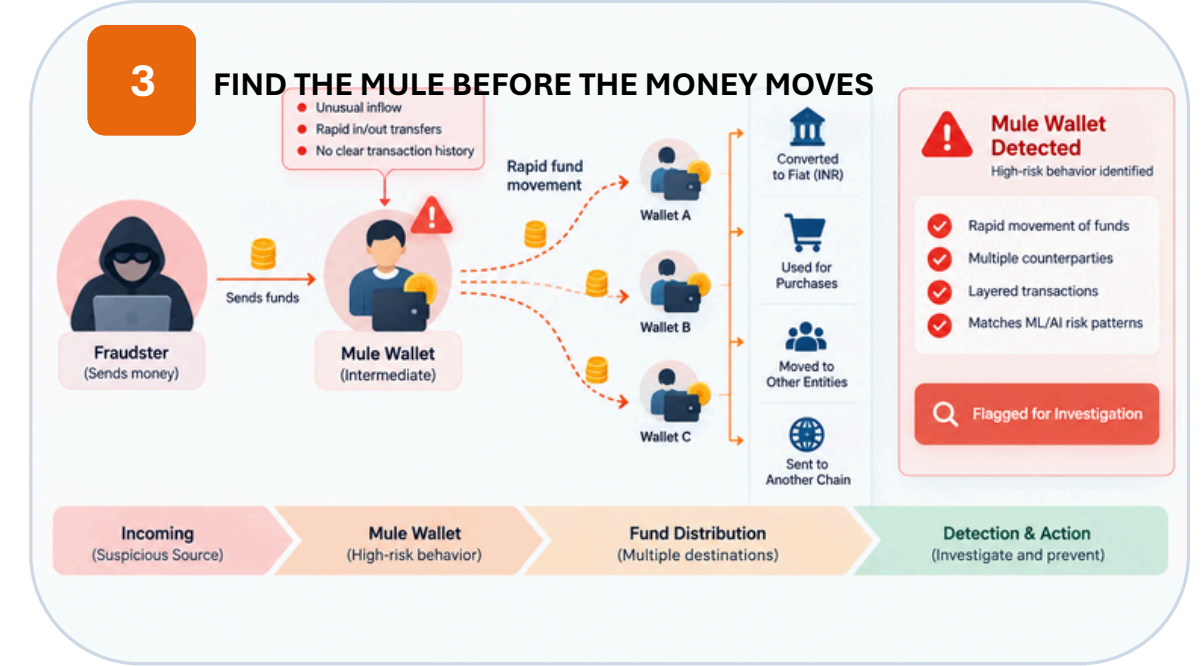
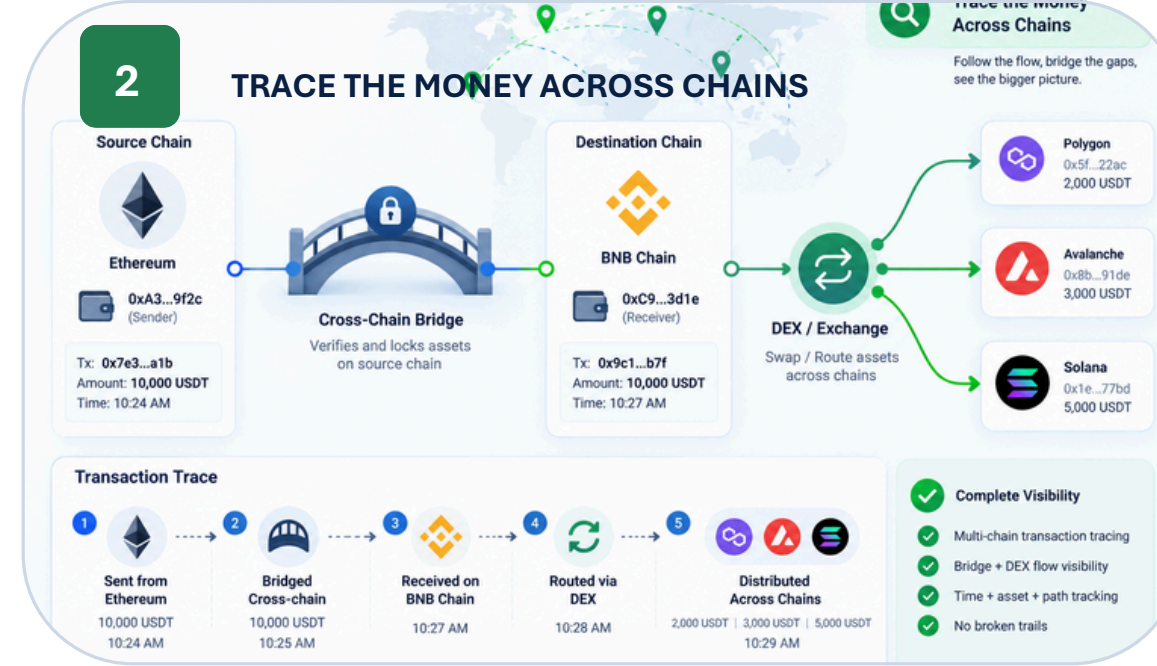


Phase 4 – National Expansion
18+ Months

- Inter-state investigations
- Cross-jurisdiction intelligence sharing



IMPACT & BENEFITS



SEARCH → CONNECT → TRACE → PREDICT → ALERT → ACT

FROM TRACE TO PREVENTION

Connect the evidence.

Detect the risk.

Predict the next move.

Alert the right institution.

Enable timely action.



REFERENECES: